Lecture 8

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Lecture 8

1. Reading Summary – Chapter 12
2. Predicting the Final Score Based on the Midterm Score
3. Marginal and Conditional Distributions
4. The Regression Line
5. More about Root Mean Square (RMS) Error
6. Example Questions
7. The Regression Effect and Regression Fallacy
8. Using Regression

- Read Chapters 13-14 for the next lecture.
- We will be transitioning to a discussion of probability.
- Quiz 4 is on Thursday at the end of lab.

Reading Summary – Chapter 12

Section 12.1: The Regression Line

- The slope of the regression line is $r * SD(y)/SD(x)$.
- The regression line goes through the point of averages.
- Don't use regression lines from observational studies to predict the results of an intervention.

Section 12.2: The Method of Least Squares

- The regression line is the line that has the smallest RMS error among all lines.

Section 12.3: Does the Regression Make Sense?

- Consider known relationships, like between area and perimeter, before blindly applying a regression.
- Other variables frequently explain the relationship between two variables of interest.

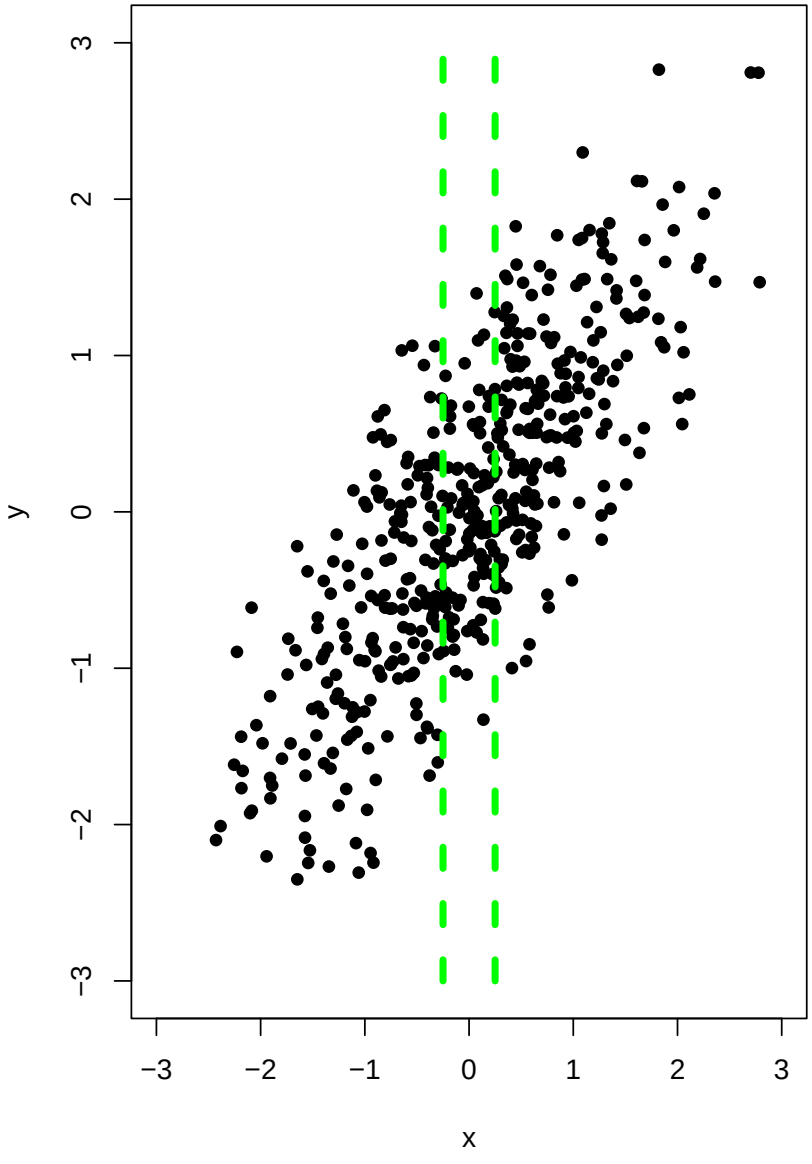
Predicting the Final Score Based on the Midterm Score

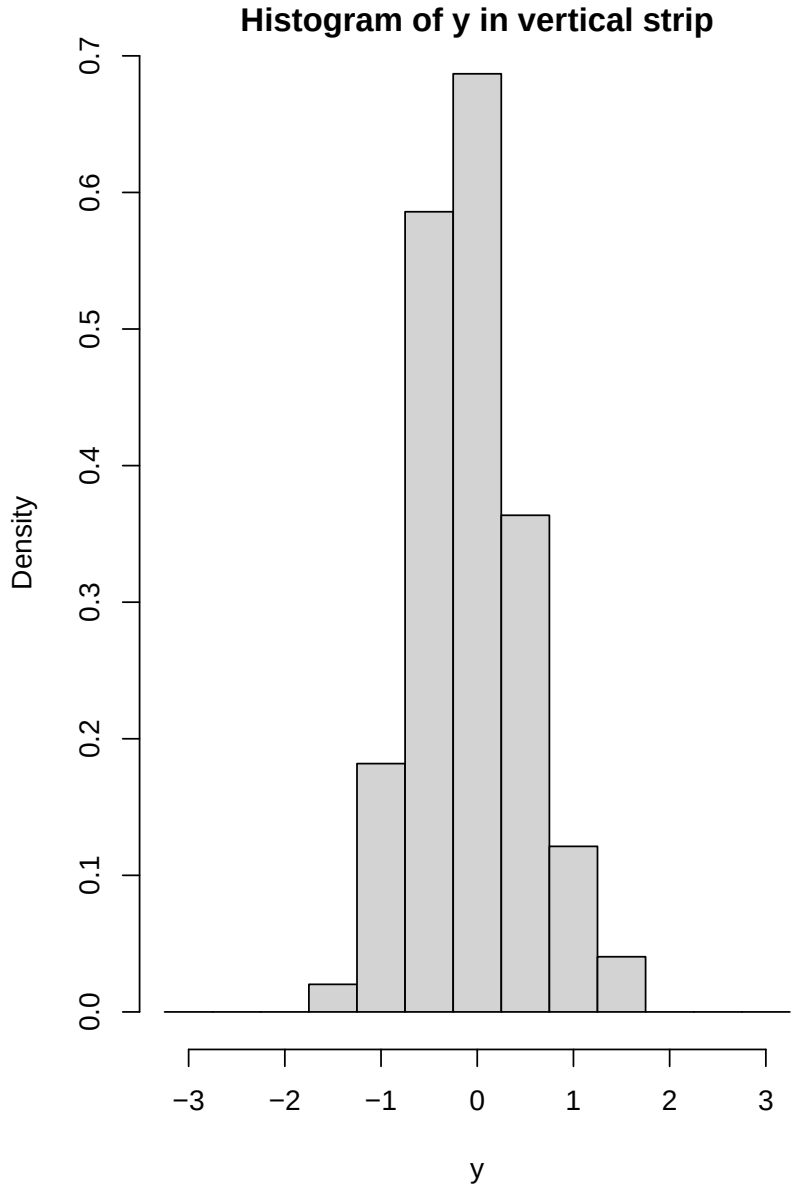
- The mean and standard deviation on the final were 75 and 15, respectively.
- The mean and standard deviation on the midterm were 60 and 15.
- The correlation between the midterm and the final scores was 0.6.
- Write an equation predicting the final score y based on the midterm score x.

Marginal and Conditional Distributions

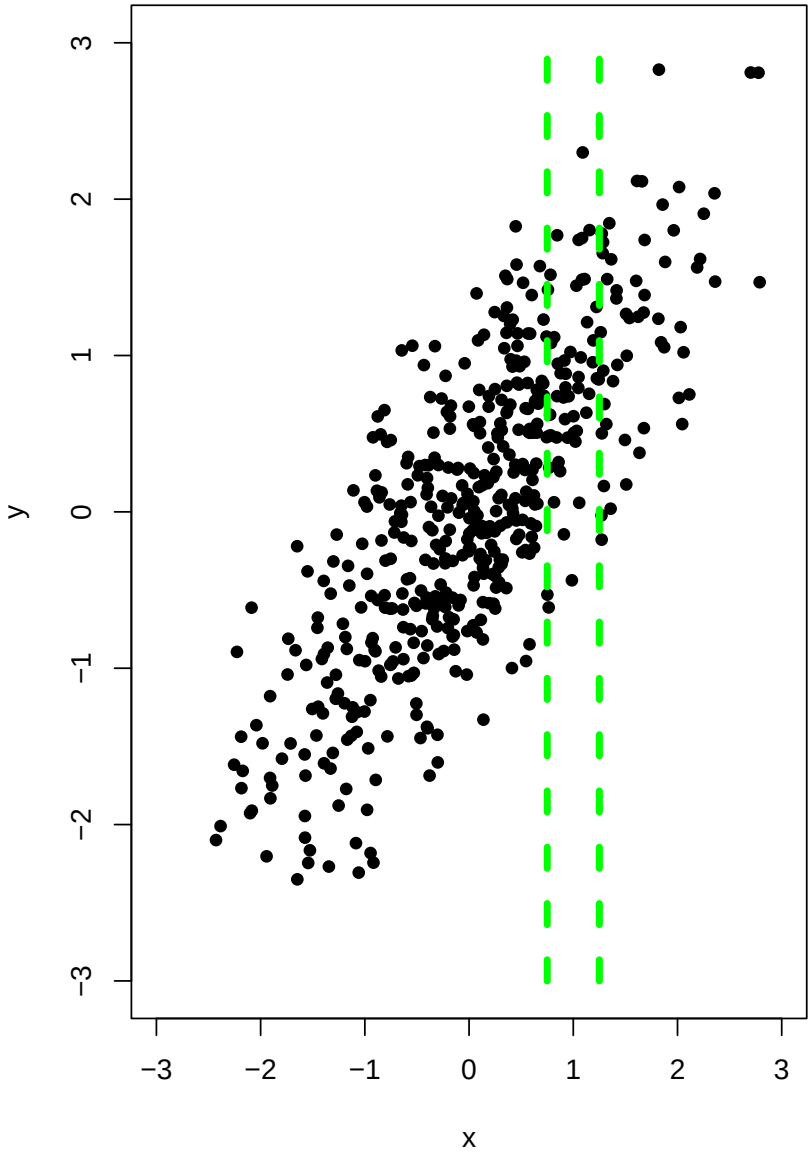
- The **conditional distribution** of y given $x \approx a$, denoted as $f(y|x \approx a)$, refers to the distribution of y in the **subset** of the data where x is approximately equal to some value a.
- In other words: the distribution of y in a vertical strip.
- The **marginal distribution** of y is the distribution of **all** of y, ignoring x.
- In other words: the distribution of y overall.

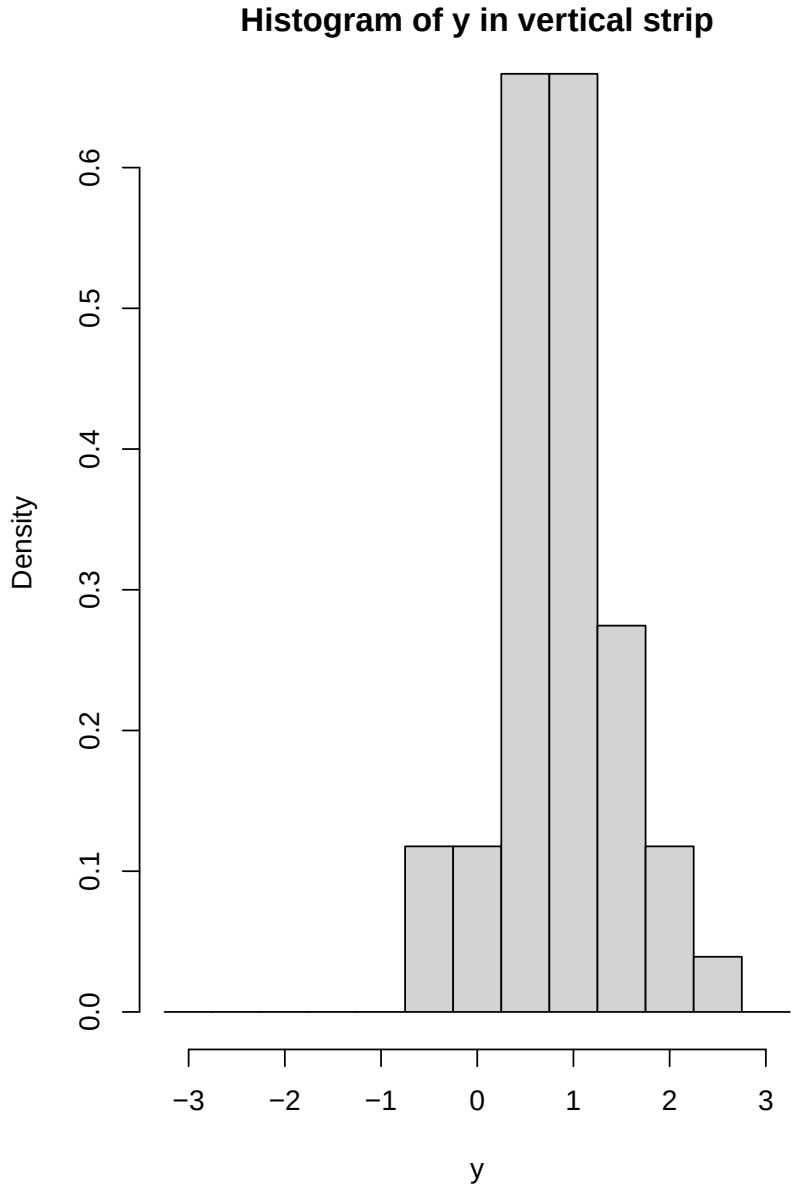
Conditional Distribution $f(y|x \approx 0)$



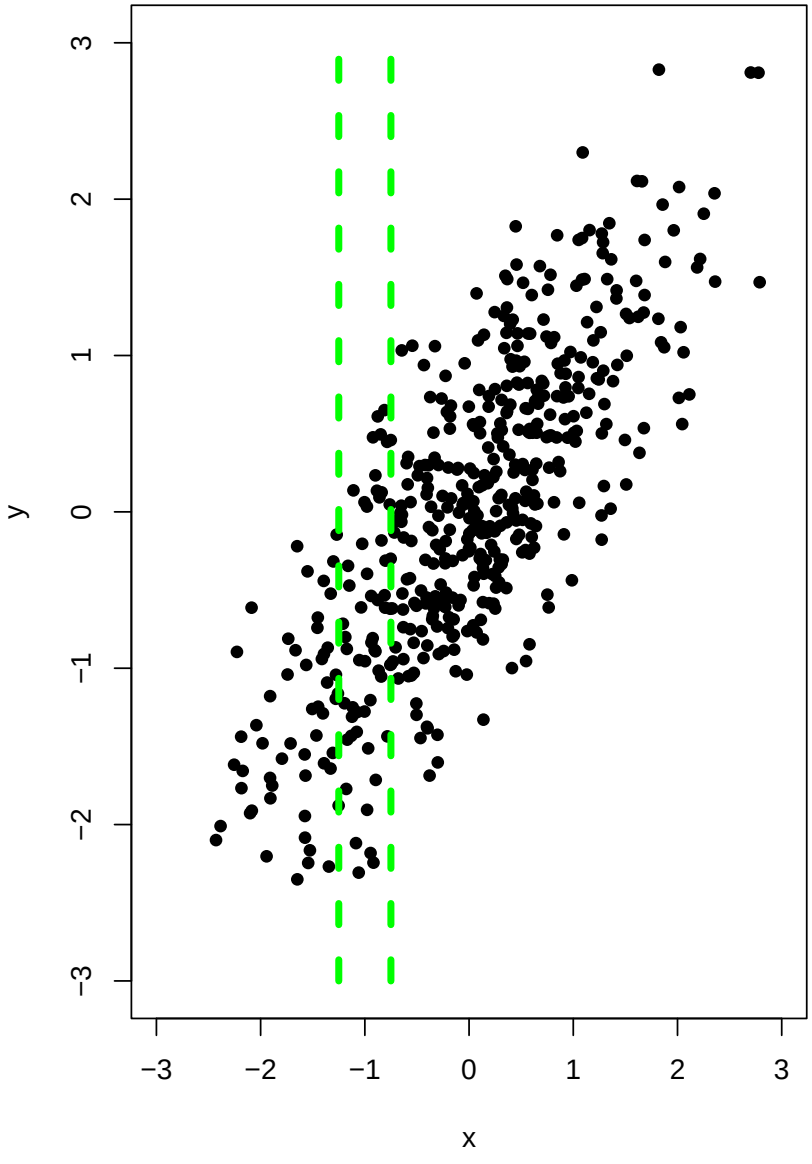


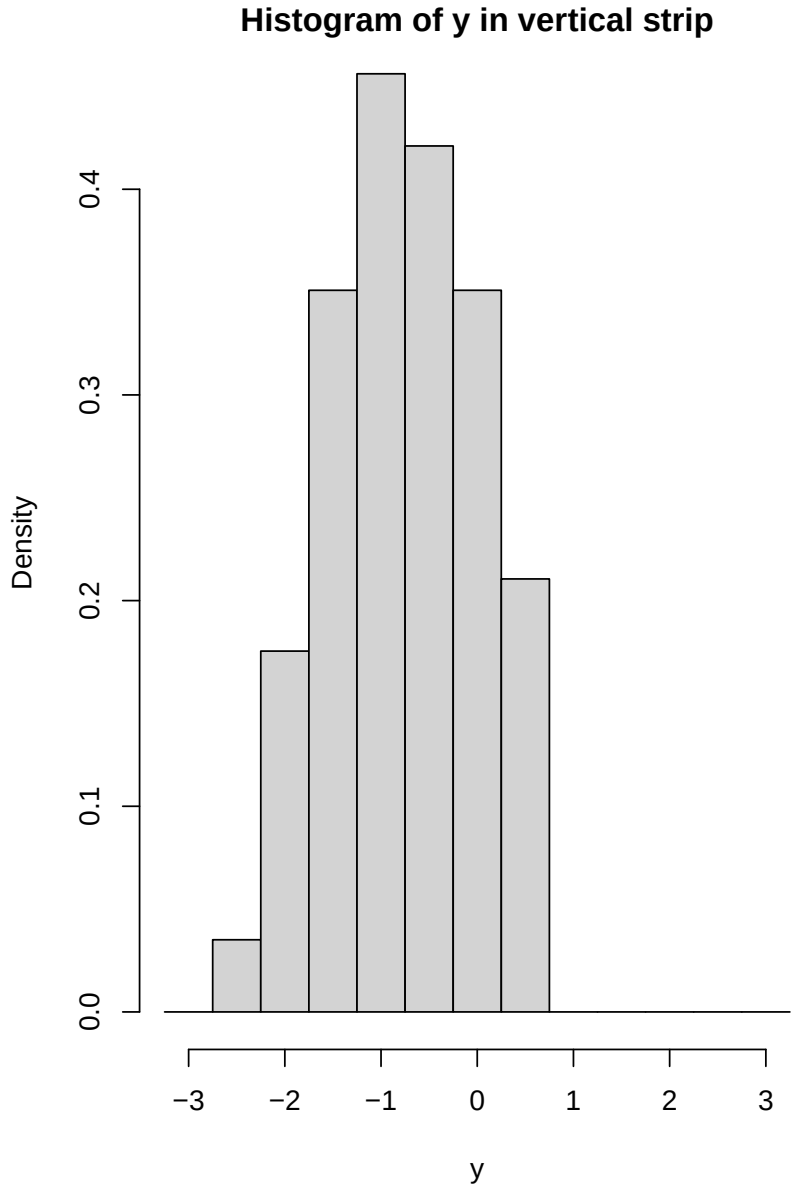
Conditional Distribution $f(y|x \approx 1)$



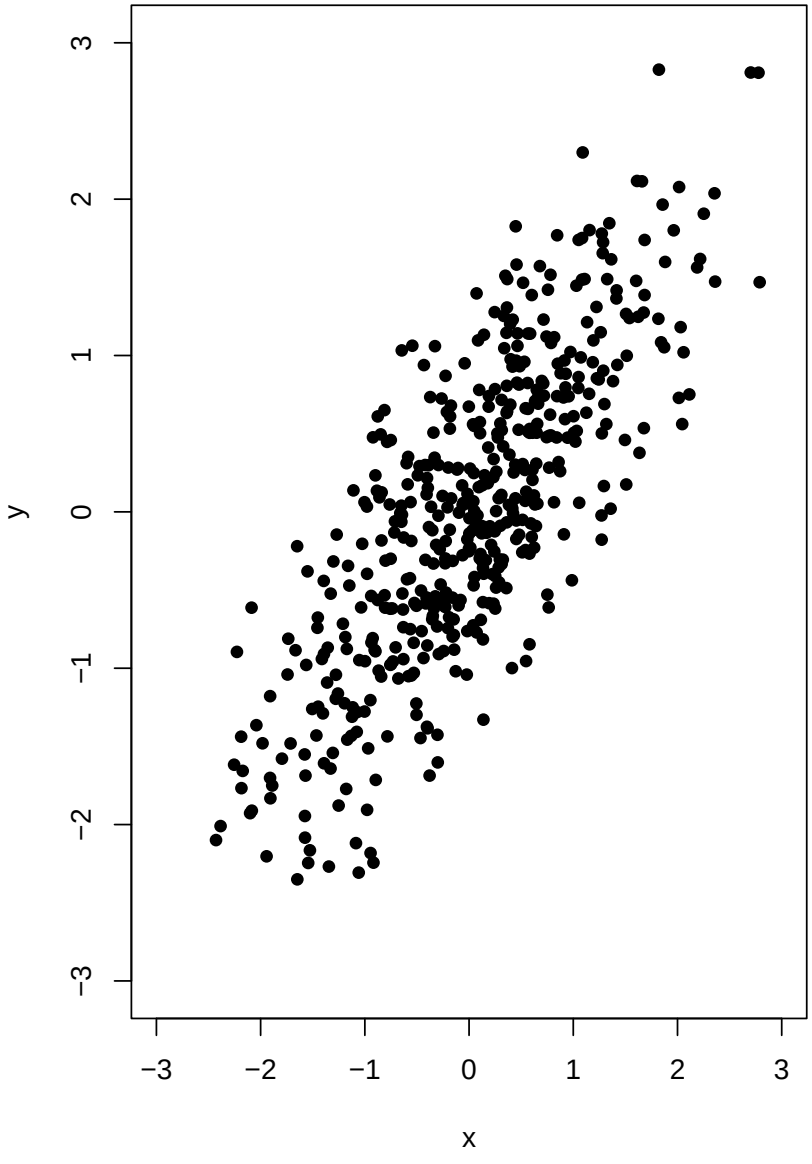


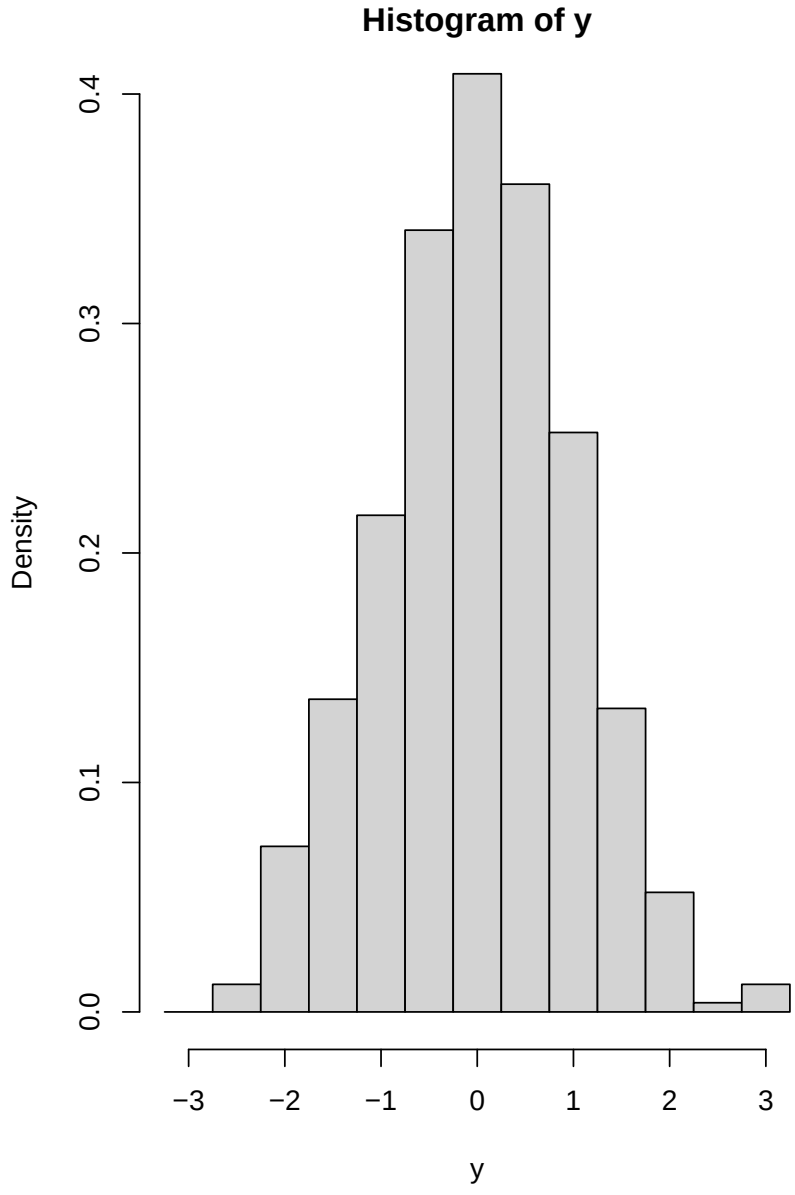
Conditional Distribution $f(y|x \approx -1)$





Marginal Distribution of y

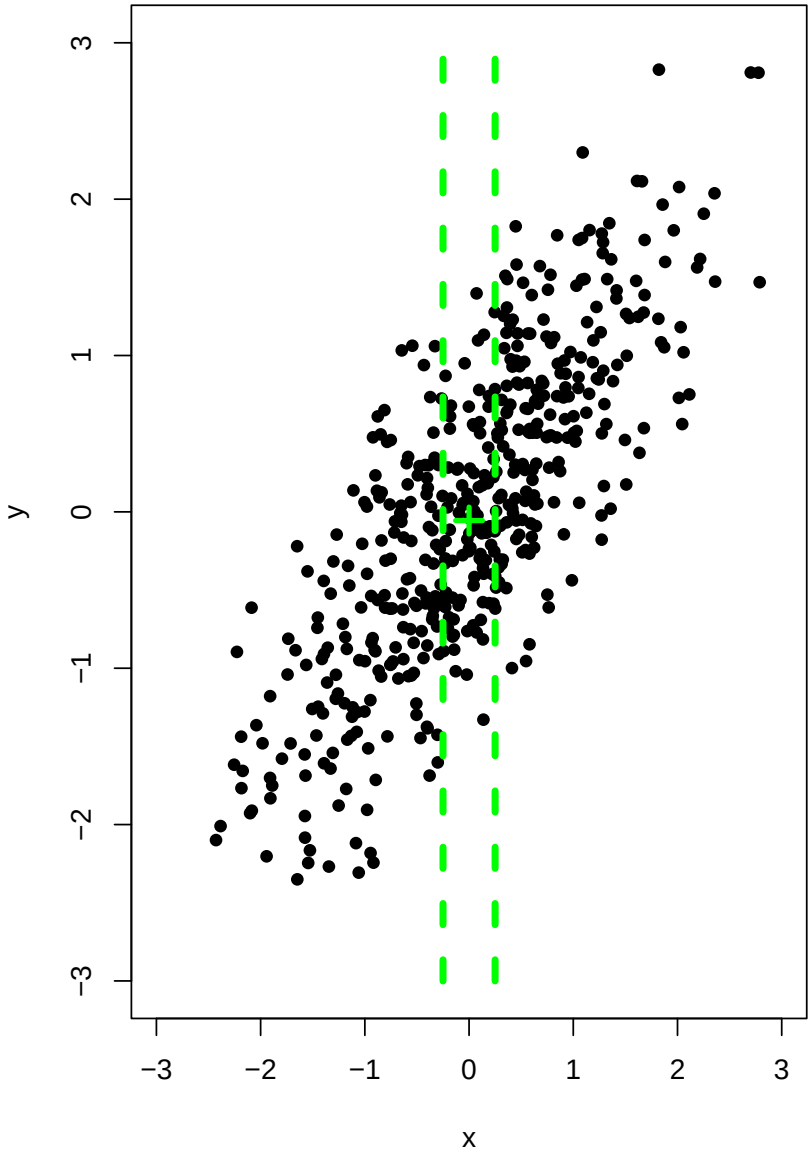


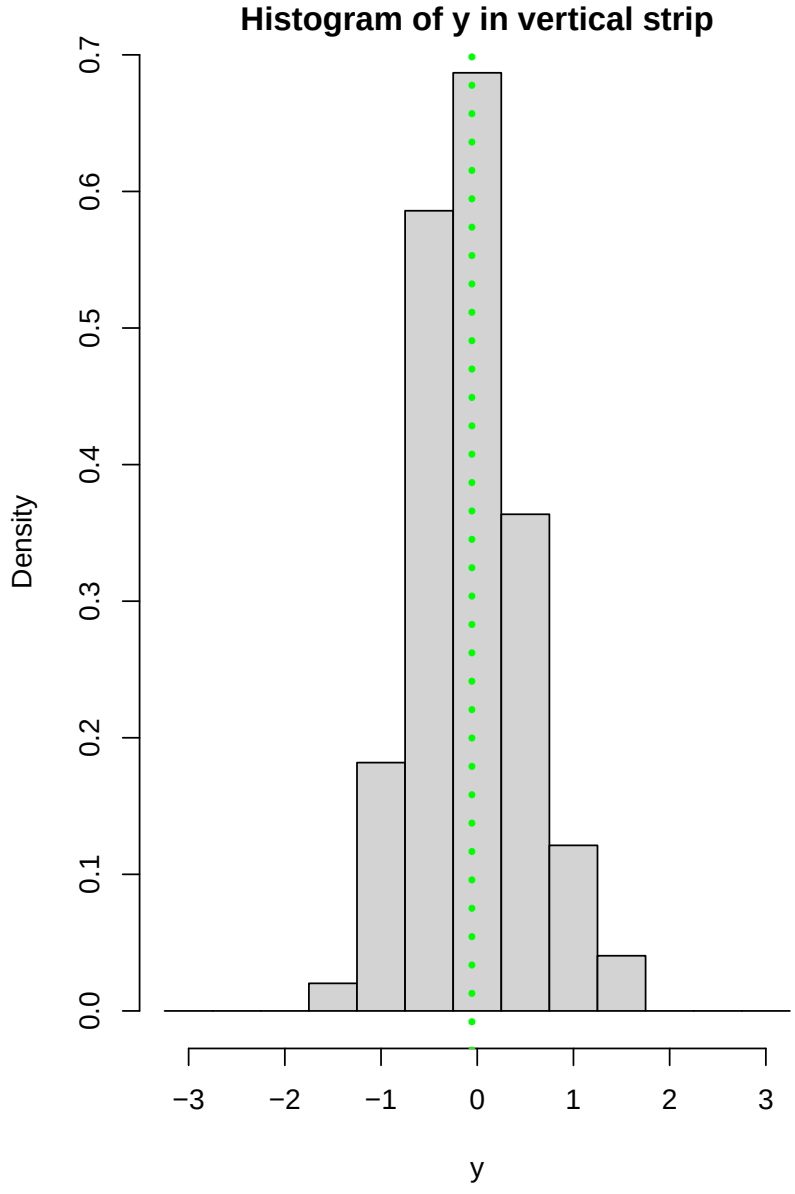


Conditional Average and Graph of Averages

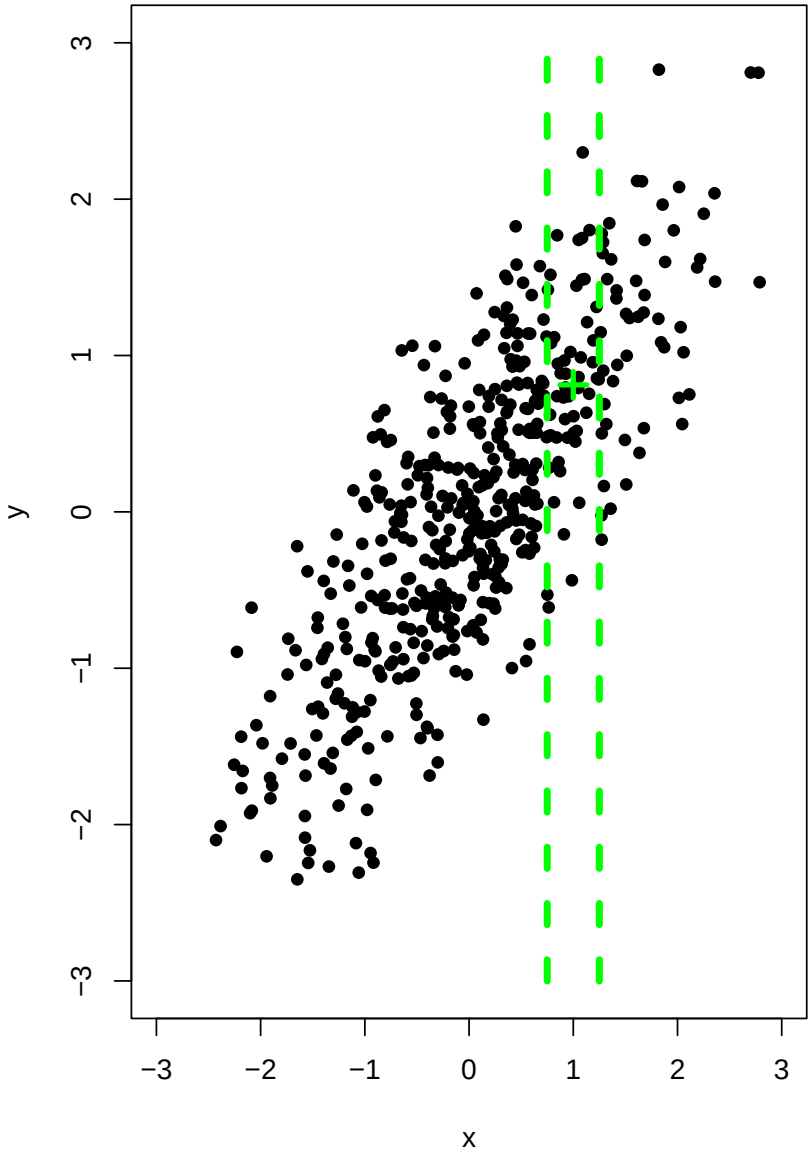
- The **conditional average** of y given $x \approx a$, denoted as $E(y|x \approx a)$, represents the average value of y within the vertical strip where x is approximately some value a.
- The **graph of averages** is the graph of $E(y|x \approx a)$ for varying values of a.

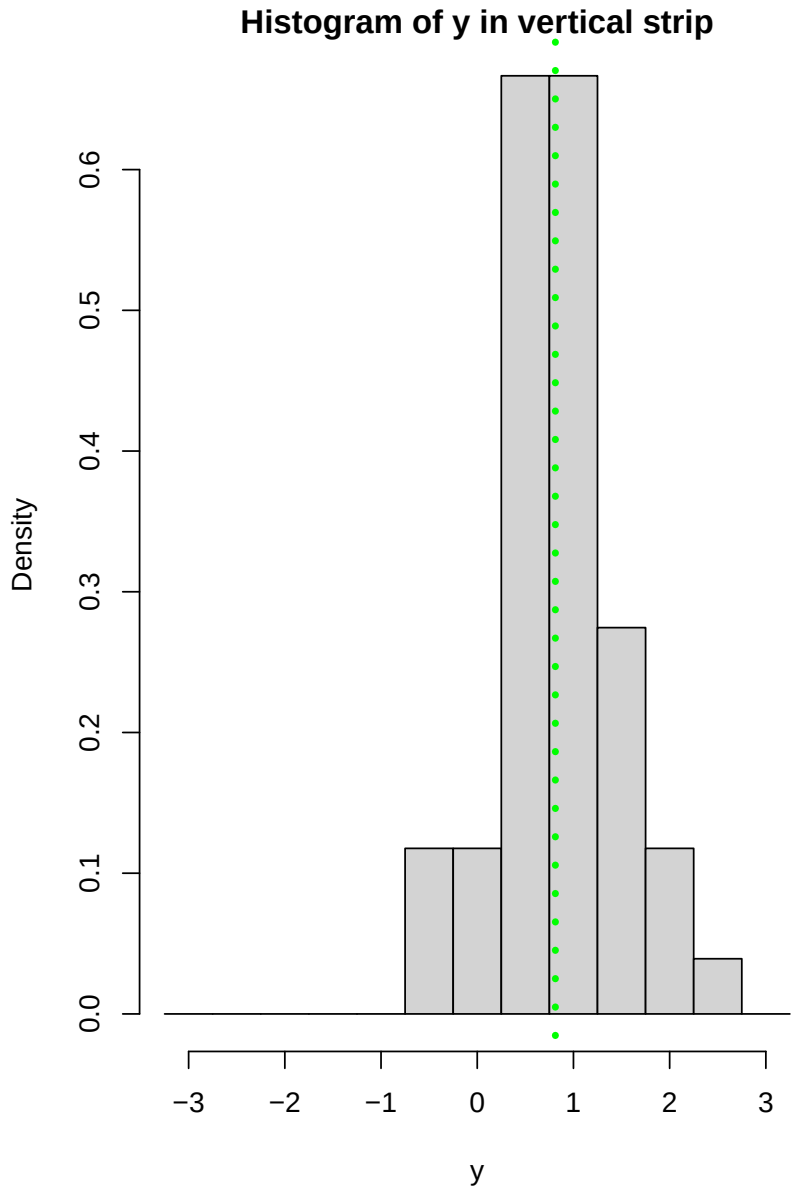
Conditional Average $E(y|x \approx 0)$



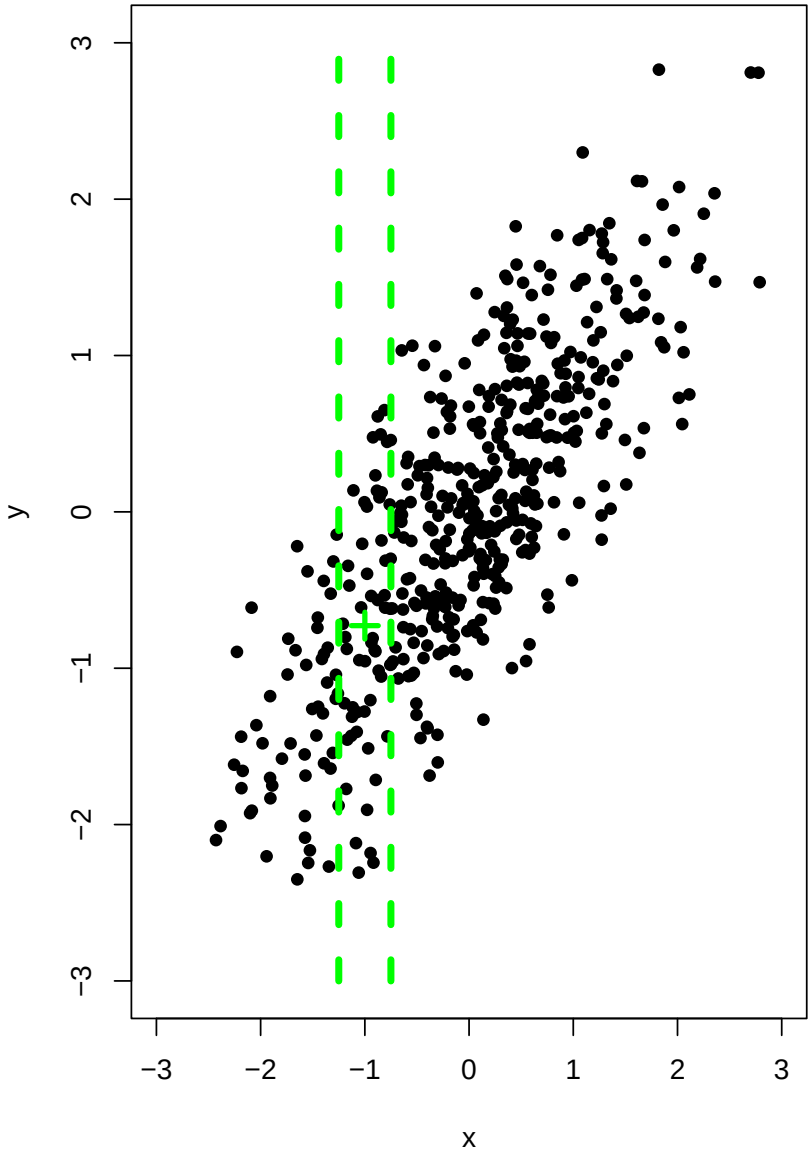


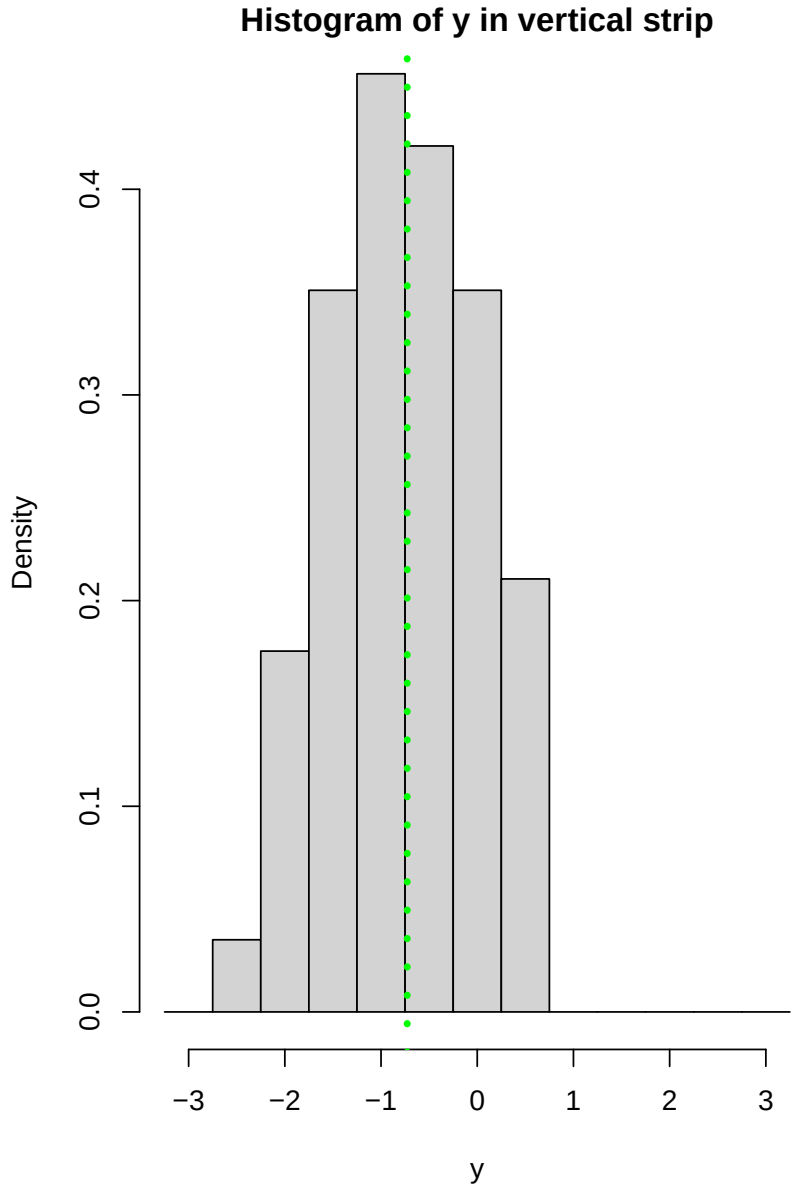
Conditional Average $E(y|x \approx 1)$



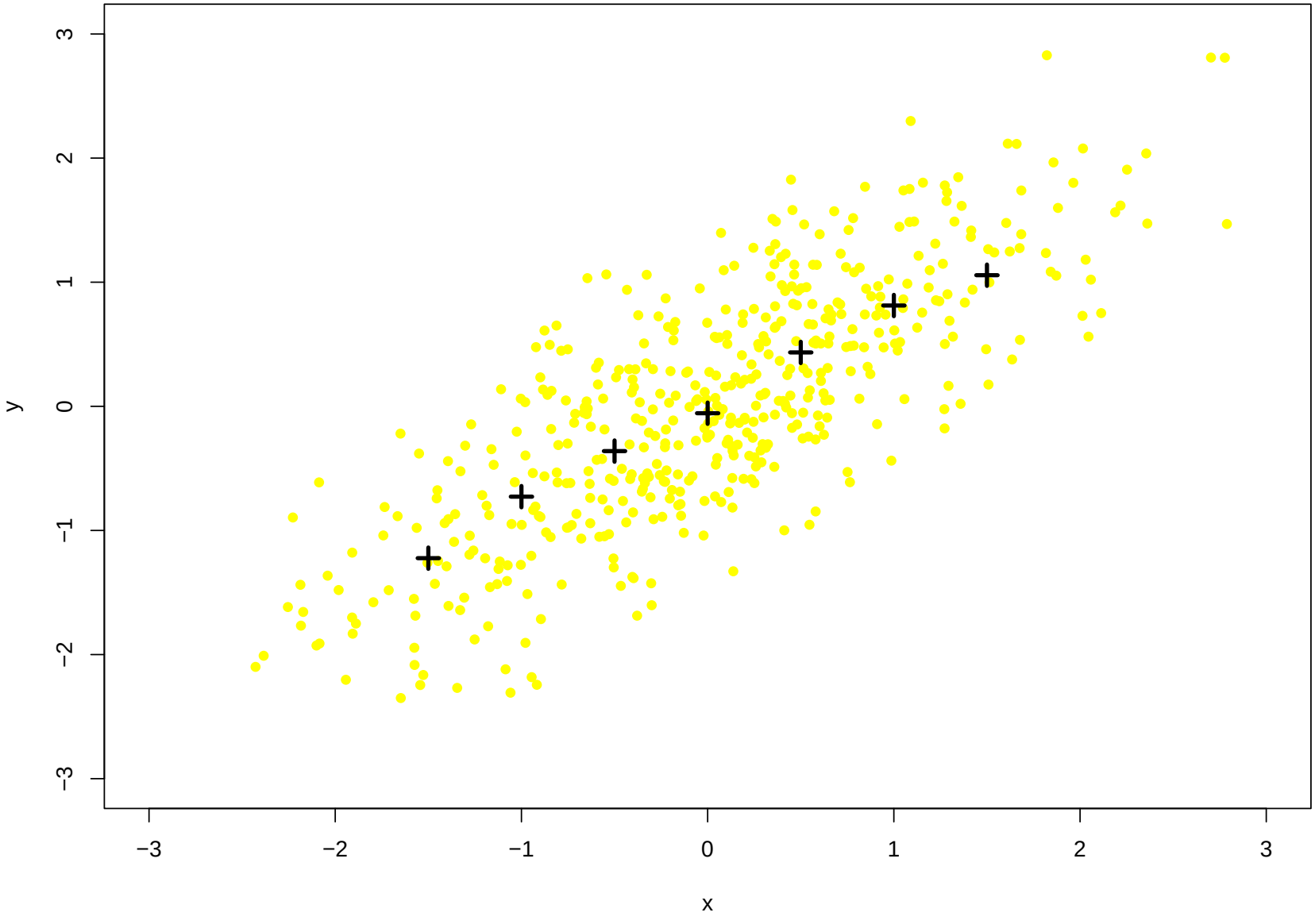


Conditional Average $E(y|x \approx -1)$



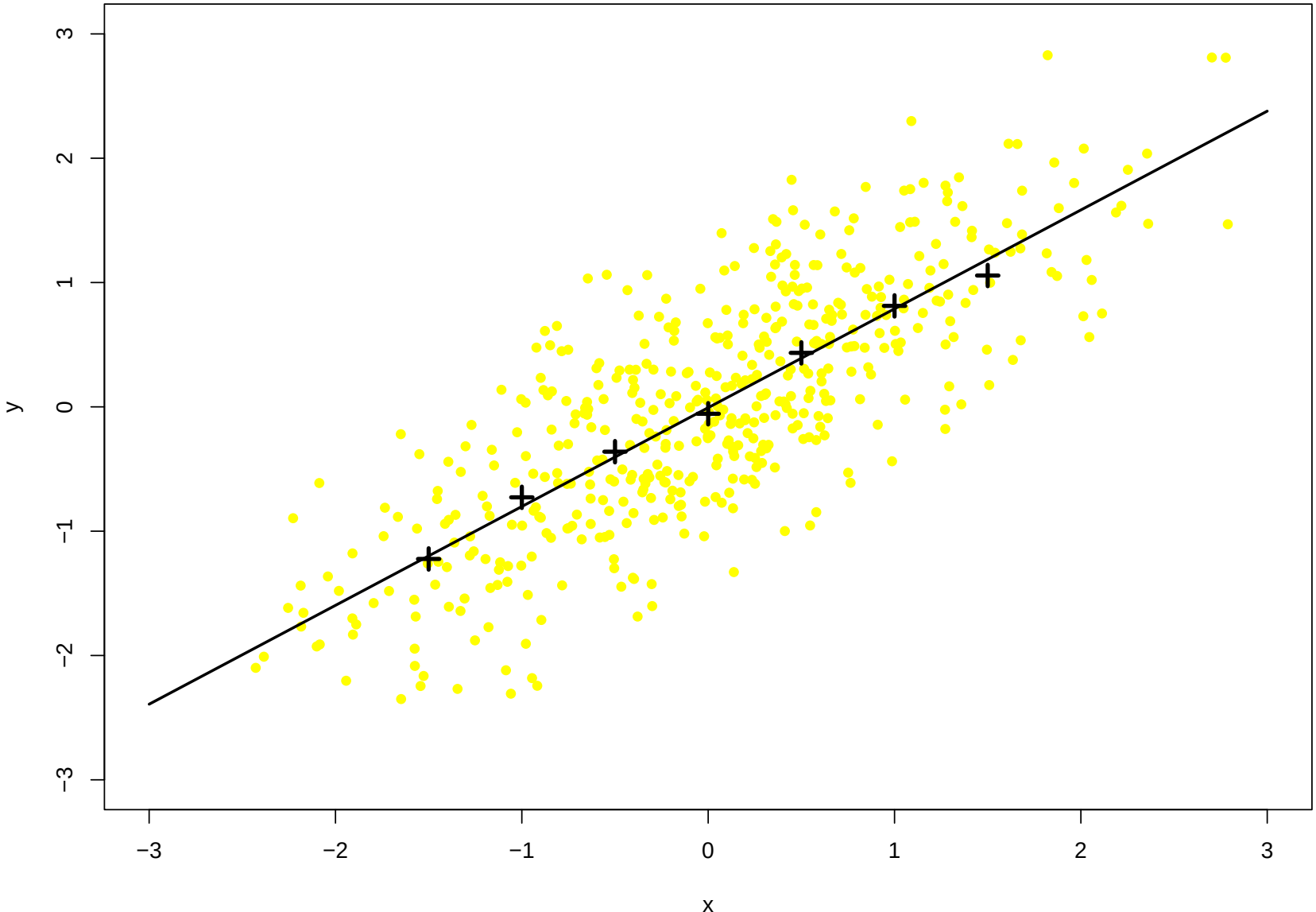


The Graph of Averages

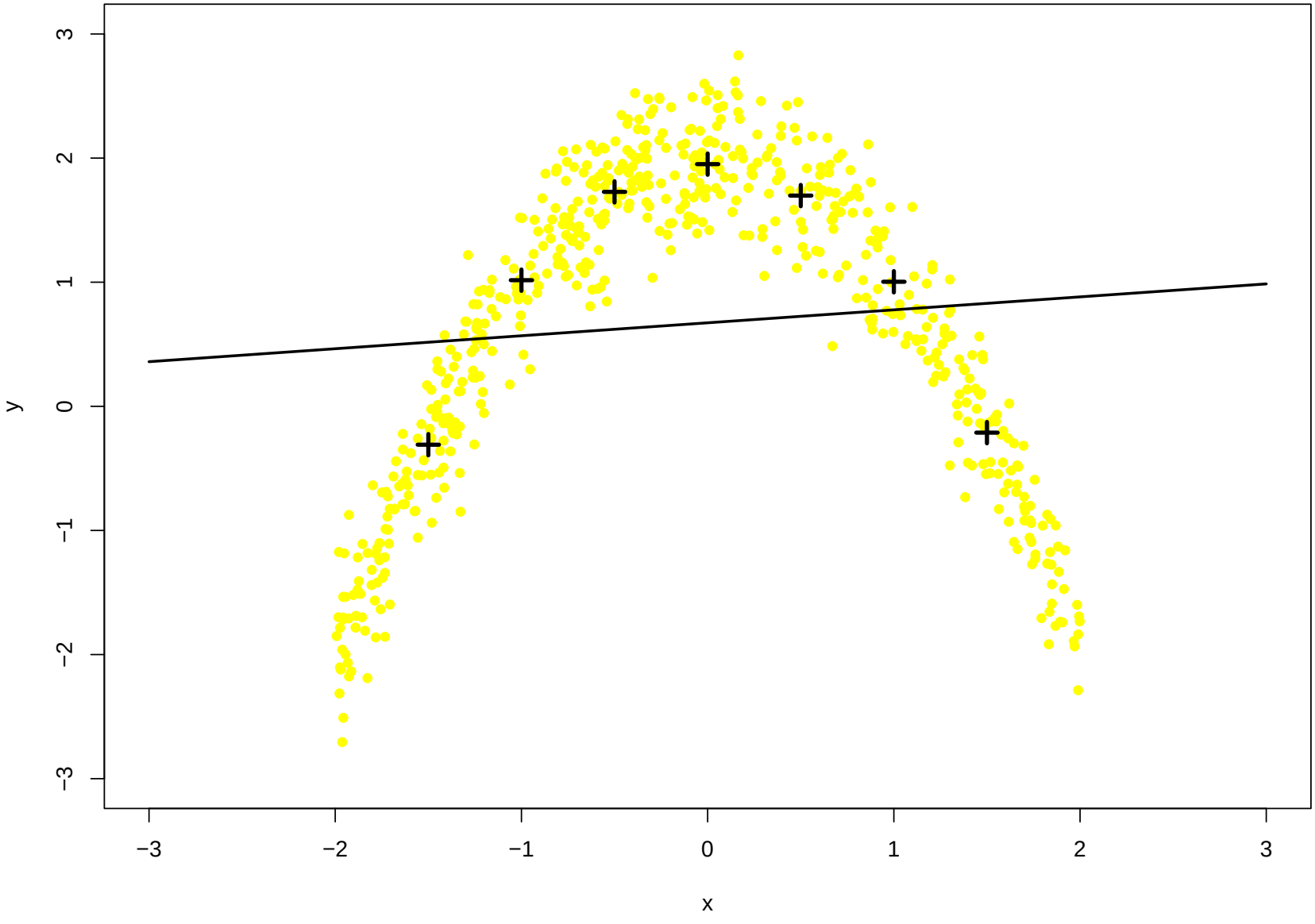


The Regression Line

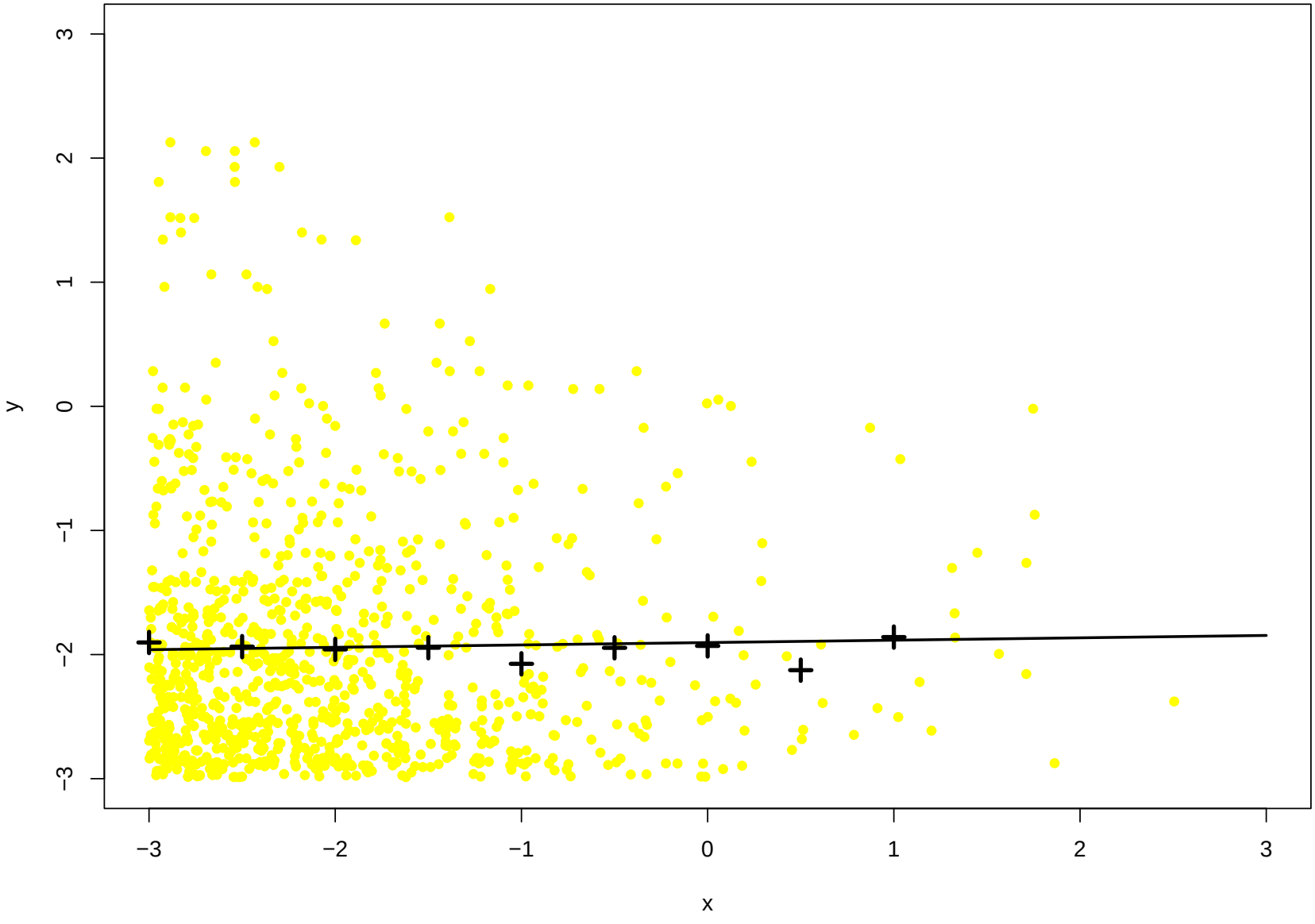
- Regression line: smoothed version of the graph of averages.
- True only when the data follow a linear trend!



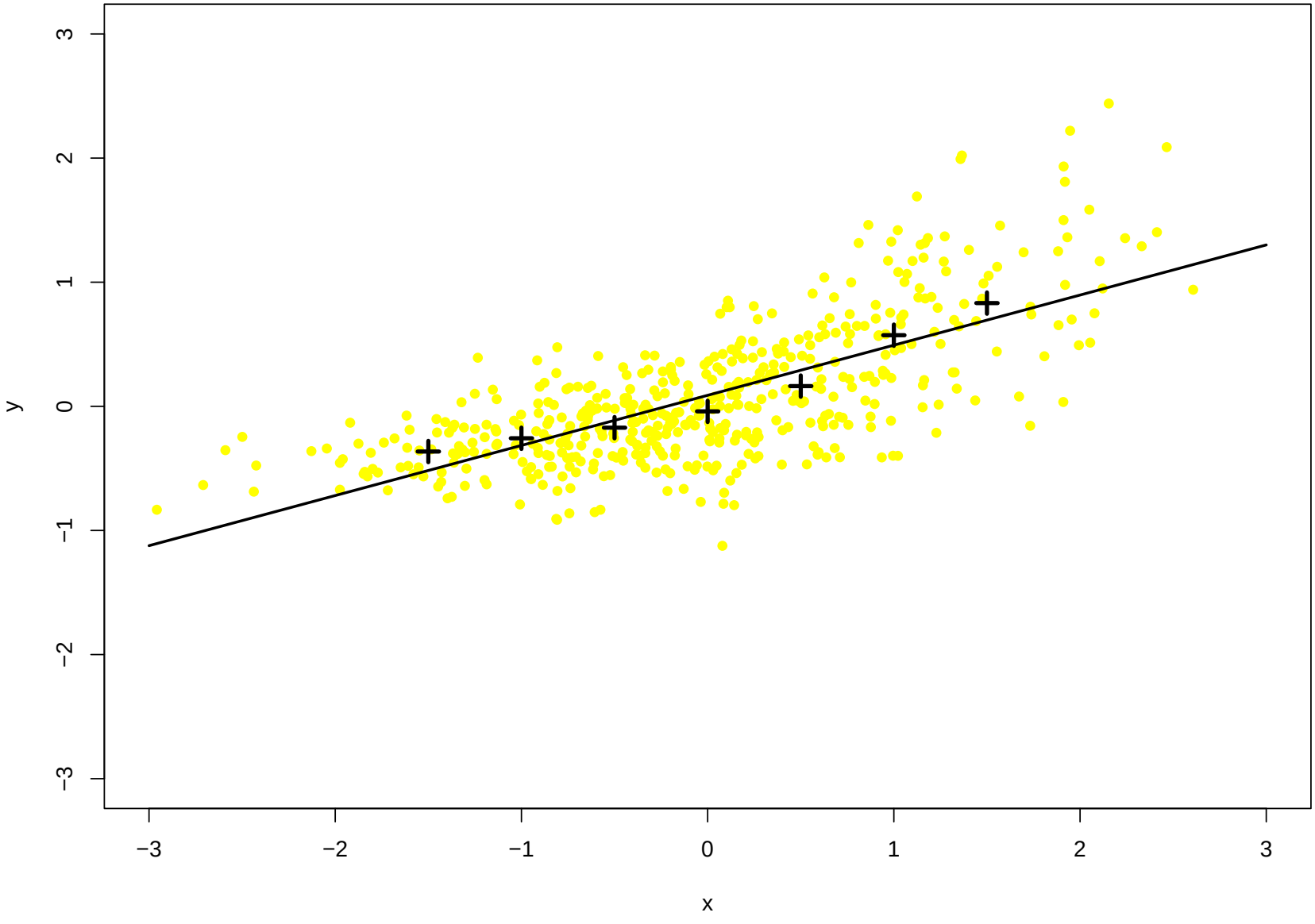
Regression Line and Non-linear Data



Regression Line and Non-linear Data Part 2



Regression Line and Non-linear Data Part 3



Where Does the Regression Line Come From?

- We know the regression line will be of the form $y = mx + b$.
- What are m and b?

Review Questions

- See Review_questions7.pdf

Least Squares

- First, define the prediction errors: $e_i = y_i - (mx_i + b)$.
- To find the regression line, we pick the m and b that minimizes the sum of squared errors $\sum_{i=1}^{n} e_i^2$.
- This quantity can be rewritten as $\sum_{i=1}^{n} [y_i - (mx_i + b)]^2$.
- This is the method of **ordinary least squares (OLS)** regression

Why Least Squares?

- Within a vertical strip, the mean minimizes squared deviations.
- If our goal is to pick a line that passes through the average of each vertical strip, then it makes sense to pick the line that minimizes the squared deviations from the line.

Solving for m and b

- After some algebra (not shown), we learn that

$$m = \frac{r \times \mathrm{SD}(y)}{\mathrm{SD}(x)}.$$

- We also know that the regression line passes through the point of averages, $(\bar{x}, \bar{y})$.
- We can use this fact to solve for b.
- Alternatively, we can write the equation of the line as $y - \bar{y} = m(x - \bar{x})$.
  - An increase in 1 SD in x is associated with only an increase of r SDs in y.

Non-linear Data

- We defined the regression line so that it would approximate the graph of averages.
- This will only work if the graph of averages actually follows a line.

Conclusion

- If the data follows a linear trend, then the regression method can be used to estimate the conditional mean of y given x.
- If the data is football-shaped, then we can accurately estimate percentiles for y given x.
- The RMS error is a good measure of this uncertainty in y for any value of x.

More About Root Mean Square (RMS) Error

- Goal: Find the SD inside a vertical strip
- The regression method allows us to find the approximate mean of y inside a vertical strip.
- If we know the SD, we could also then use the normal approximation to calculate percentiles, etc.

Note: Everything here, and in the next few slides, assumes the data is football shaped.

SD in a Vertical Strip

- Recall: The SD is the square root of the average of the squared deviations from the mean.
- Recall: The average inside a vertical strip is approximated by the regression line.
- Thus: Inside a vertical strip, the SD is approximately equal to the square root of the average of the squared deviations from the regression line.
- Deviations from the regression line are the same as prediction errors: actual value - predicted.

Football Shaped Data is Homoscedastic

- When the data is football shaped, the SD is approximately the same in every vertical strip.
- In other words, the average of the squared deviations from the regression line is approximately the same within every vertical strip.
- Conclusion: Don't bother with individual vertical strips.
  - Just calculate the square root of the average of all squared deviations from the regression line.

The RMS Error

- Let e_i be the deviation (prediction error) of the i-th data point.
- The RMS error is defined as:

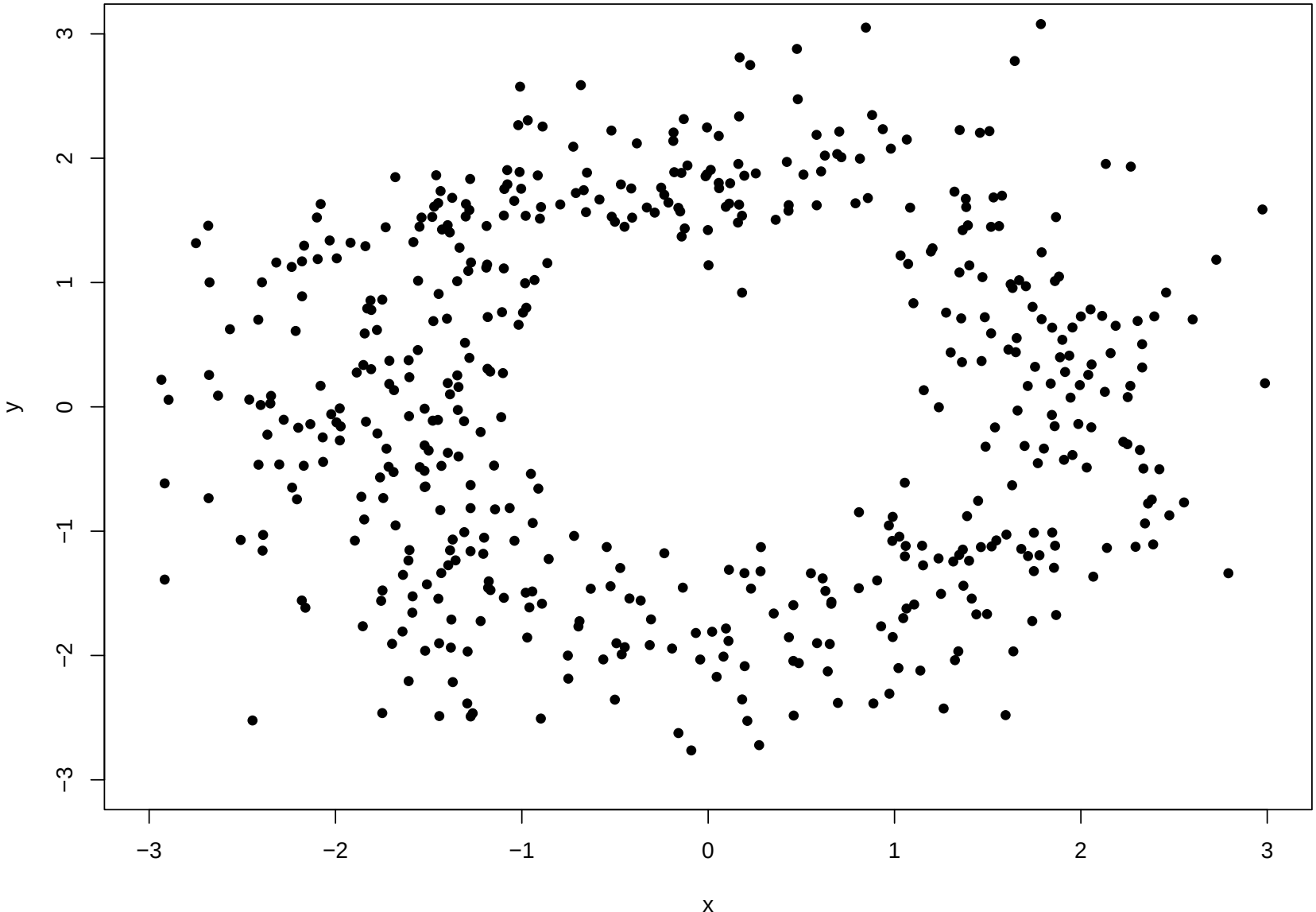
$$\text{RMS error} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} e_i^2}$$

- When the data is football shaped,

$$\text{SD in a vertical strip} \approx \text{RMS error}$$

Discussion Question

- Estimate the SD of a vertical strip using the RMS error.
- Is your estimate too big, too small, or about right?



Algebraic Fact

- The formula below provides a convenient way of calculation the RMS error:

$$\sqrt{\frac{1}{n}\sum_{i=1}^{n} e_i^2} = \sqrt{1-r^2} \times \mathrm{SD}(y).$$

- This equality is true just by algebra. It doesn't require any special assumptions.

Discussion Questions

- Which of the following are true:
  1) For any dataset, football shaped or not, the RMS error is equal to $\sqrt{1-r^2} \times \mathrm{SD}(y)$
  2) For any dataset, $\sqrt{1-r^2} \leq 1$
  3) For any dataset, the RMS error is less than or equal to the $\mathrm{SD}(y)$.
  4) For any dataset, the SD inside any vertical strip is less than or equal to the overall $\mathrm{SD}(y)$.
  5) For football shaped data, the SD inside any vertical strip is less than or equal to the overall $\mathrm{SD}(y)$.

Example Questions

- A statistical analysis was made of the midterm and final scores in a large course, with the following results:
  - Midterm: Avg $= 50$, SD $= 25$
  - Final: Avg $= 55$, SD $= 15$
  - $r = 0.6$
  - The scatter diagram is football-shaped.
  - We used the regression line to predict the final score based on the midterm score for each student.

Example Questions (continued)

1. Predict the final score for a student whose midterm score was 80
2. This prediction is likely to be off by about how many points?
3. For about 5% of the students, the prediction for the final score was off by more than _____ points.
4. About what percentage of the students scored over 80 on the final?

Example Questions Part 3

5. Of the students who scored 80 on the midterm, about what percentage scored over 80 on the final?

6. Of the students who scored at the 70th percentile on the midterm, about what percentage scored over 80 on the final?

7. Of the students who scored at the 70th percentile on the midterm, about what percentage scored above the 70th percentile on the final?

Solutions to Example Questions

1. $z = (80 - 50)/25 = 1.2, z_{final} = 1.2 * 0.6 = 0.72, pred = 55 + 0.72 * 15 = 65.8$
2. $RMS = \sqrt{1 - 0.6^2} * 15 = 12$
3. $2 * 12 = 24$
4. $z = (80 - 55)/15 = 1.7, (100\% - 91\%)/2 = 4.5\%$ in right tail
5. Predicted value 65.8 from 1., RMS 12 from 2.,
   $z = (80 - 65.8)/12 = 1.2$, $(100\% - 77\%)/2 = 11.5\%$

Solutions to Example Questions (continued)

6. $70\% - (100\% - 70\%) = 40\%$ of area in middle, $z = 0.5$,
$z_{final} = 0.5 * 0.6 = 0.3, score = 55 + 0.3 * 15 = 59.5$,
$z = (80 - 59.5)/12 = 1.7, (100\% - 91\%)/2 = 4.5\%$ in right tail.

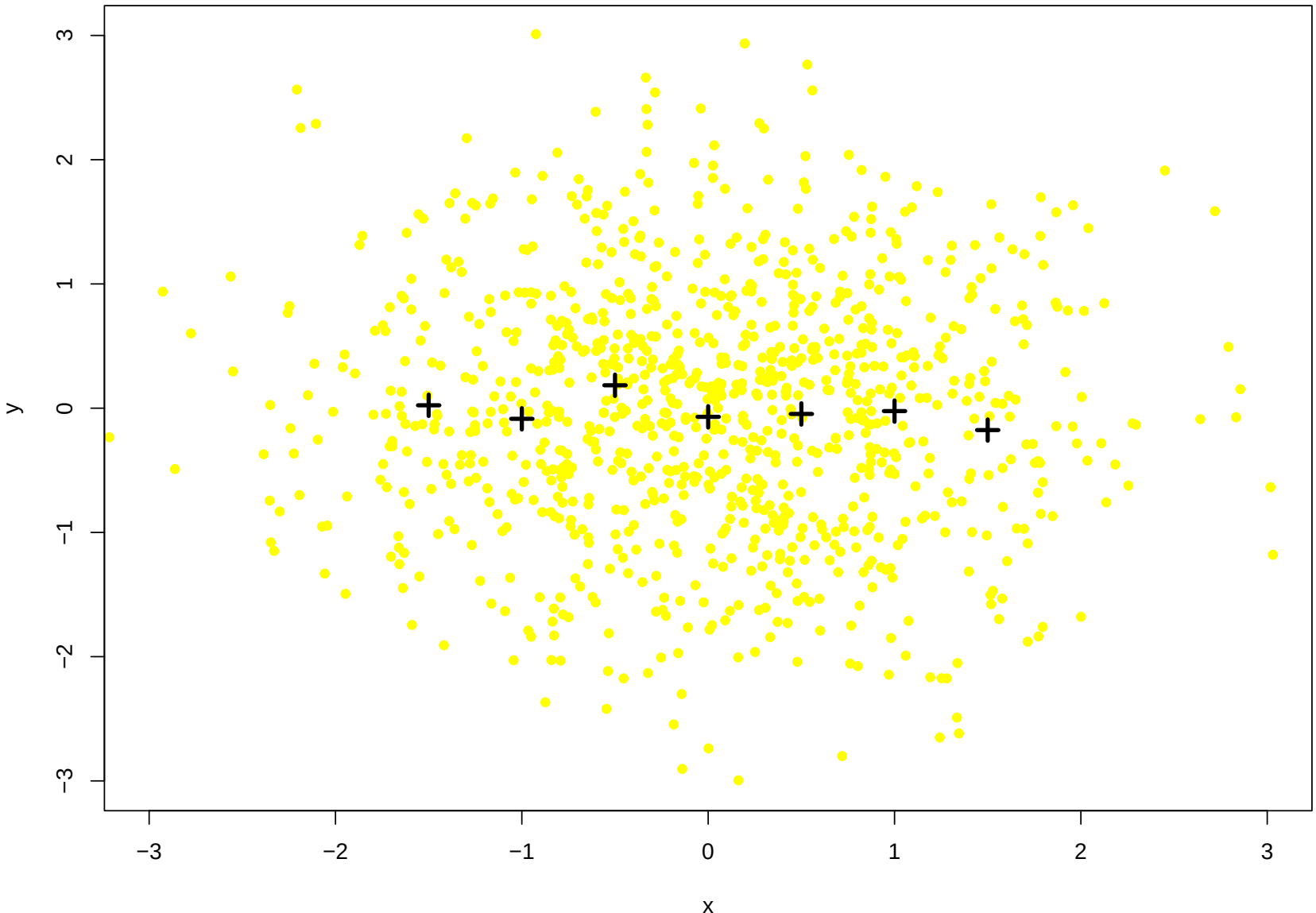
7. 70th percentile on final, $z = 0.5$ from 6.,
$score = 55 + 0.5 * 15 = 62.5$, $(62.5 - 59.5)/12 = 0.25$,
$(100\% - 20\%)/2 = 40\%$ in right tail.

The Regression Effect and Regression Fallacy

- In virtually all test-retest situations, the bottom group on the first test will on average show some improvement on the second test, and the top group will on average fall back.
- Why?

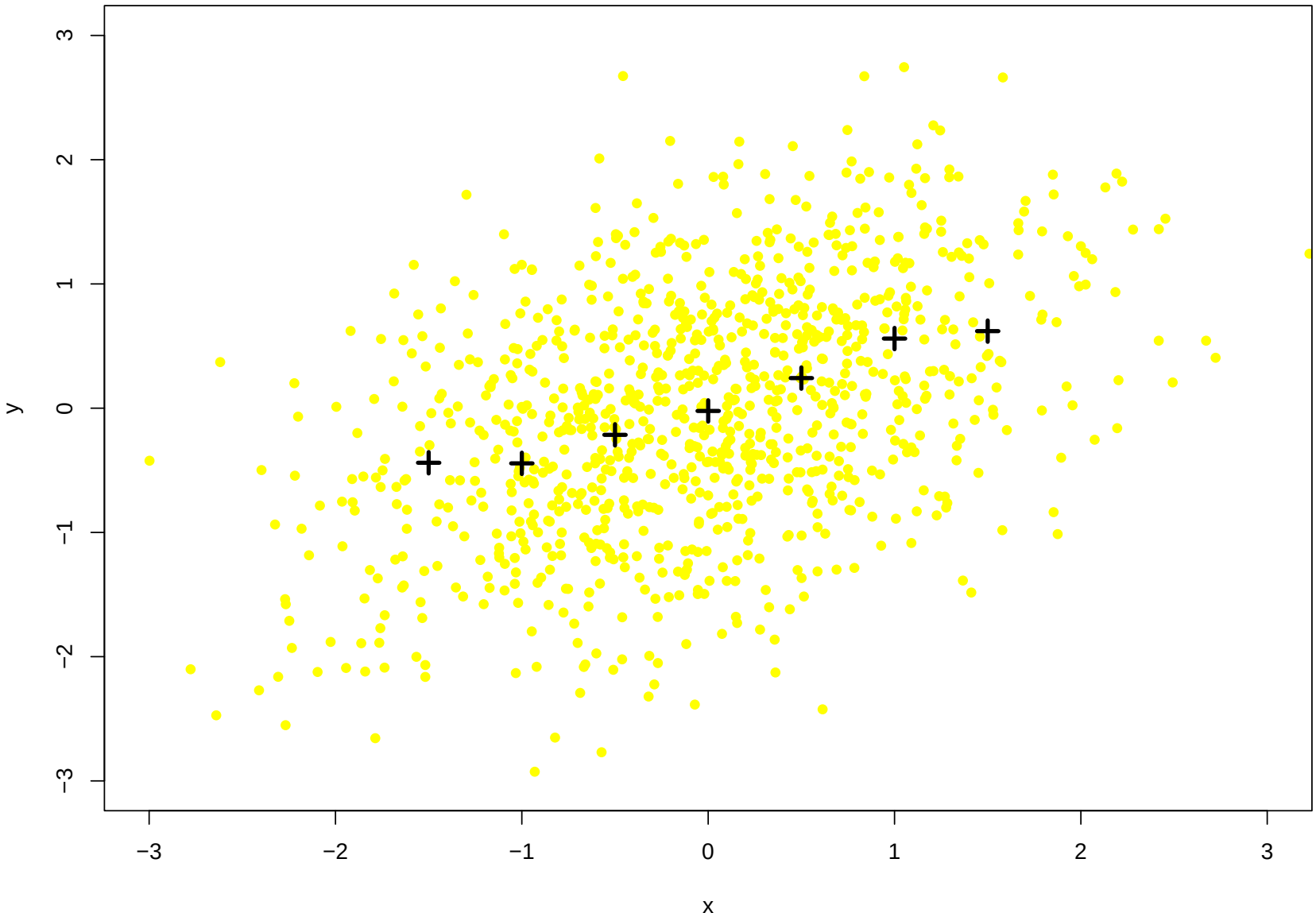
Thought Experiment

- Suppose one test is on math, the other on history.
- What would the scatter plot and graph of averages look like?



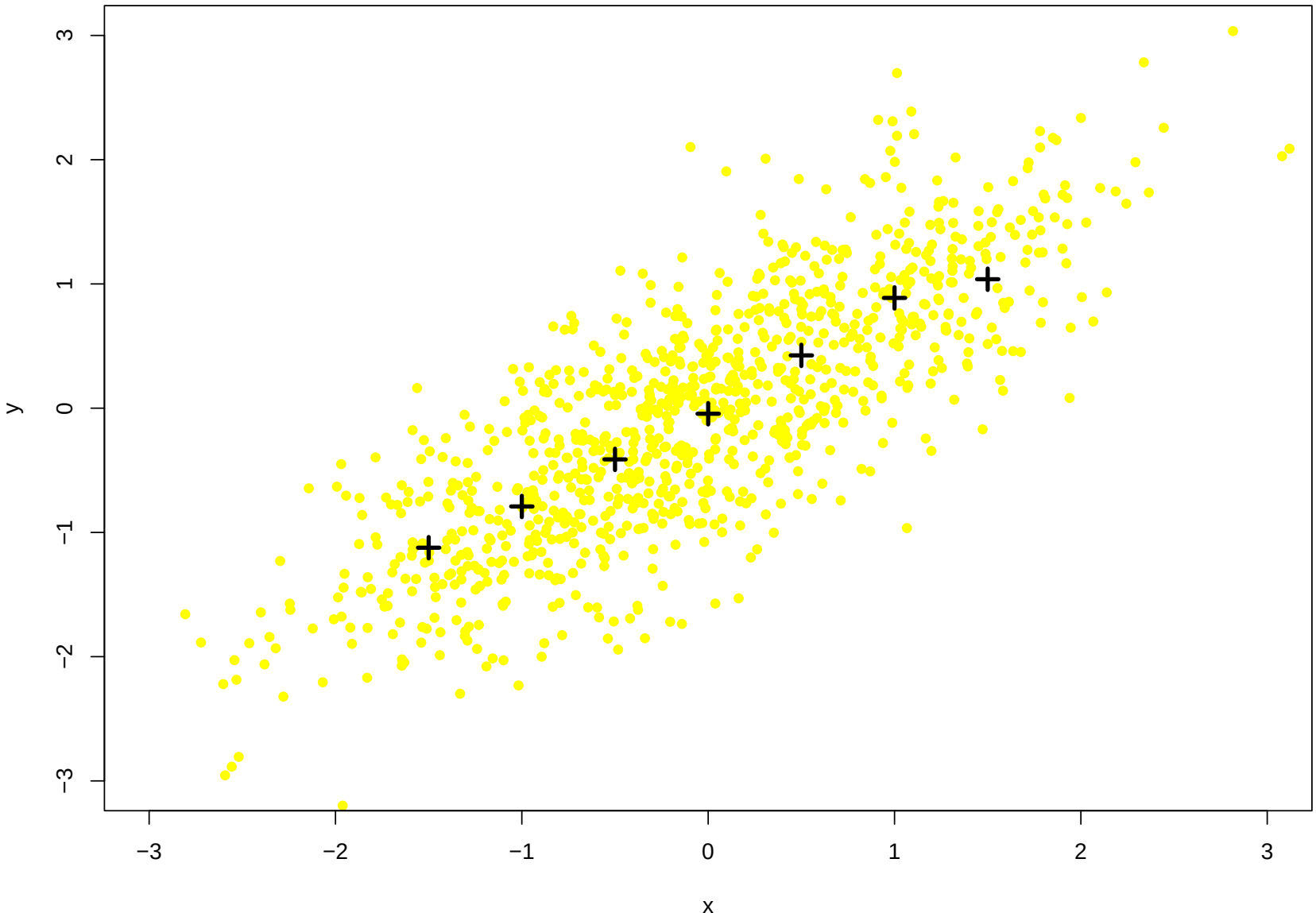
Thought Experiment Part 2

- Suppose one test is on math, the other on physics.
- What would the scatter plot and graph of averages look like



Thought Experiment Part 3

- Suppose the two tests are on the same topic.
- What would the scatter plot and graph of averages look like?



Regression Effect Conclusion

- In virtually all test-retest situations, the bottom group on the first test will on average show some improvement on the second test, and the top group will on average fall back.
- Why?

The Regression Fallacy

- It's easy to mistake the regression effect as something interesting.
- Example: Suppose a teacher takes the lowest 10 performing students from the midterm, and provides them with extra tutoring.
- On the final, their scores improve on average.
- Does this indicate the tutoring helped?
  - Not necessarily. It could very easily be nothing more than the regression effect.
  - How could you design a study to account for the regression effect?

Using Regression

The Regression Method

- **If the data follow a linear trend**, then:
- To estimate the average value of y for a given value of x without calculating the equation for the regression line,
  1. Convert x to standard units.
  2. Multiply by r. This gives you the average value of y in standard units.
  3. Convert from standard units back to the original scale for y.
- The same procedure works for percentiles:
  - Convert to and from standard units using the normal table.

Example 1

- In a certain class, midterm scores average out to 60 with an SD of 15, as do the scores for the final.
- The correlation between midterm scores and final scores is about 0.5.
- The data are football shaped.
- Predict the final score for a student whose midterm score was:
  - 75
  - 30
  - 60
  - unknown

Example 1 (continued)

- What is the root mean squared error for predicting the scores on the final based on the scores on the midterm?
  - How does this compare to the error for predicting the score on the final without using the score on the midterm?
- If the midterm score was 60, then what range would we expect 95% of the final scores to be in?

Example 2

- For the first-year students at a certain university, the correlation between SAT scores and first-year GPA was 0.7.
- The scatter diagram is football-shaped.
- Predict the percentile rank on the first-year GPA for a student whose percentile rank on the SAT was
  - 90%
  - 30%
  - 50%
  - unknown

Solutions to Example 1

1. $z = (75 - 60)/15 = 1, z_{final} = 1 * 0.5 = 0.5, pred = 60 + 0.5 * 15 = 67.5$
2. $z = (30 - 60)/15 = -2, z_{final} = -2 * 0.5 = -1, pred = 60 - 1 * *15 = 45$
3. If they score at the mean on the midterm, you would predict that they would score at the mean, or 60, on the final.
4. If you do not know the score on the midterm, then you would predict the mean score for the final, or 60.
5. The RMS error is $\sqrt{1 - 0.5^2} * 15 = 13$.
6. $60 \pm 2 * 13 = 34 - 86$.

Solutions to Example 2

1. The 90th percentile corresponds to 80% of the area in the middle, and the table in the book gives $z = 1.28$. The predicted z-score on the final would be $0.5 * 1.28 = 0.64$, and the percentile ranking is roughly $48\% + (100\% - 48\%)/2 = 74\%$.
2. The 30th percentile corresponds to 40% of the area in the middle, and the table in the book gives $z = -0.52$. The predicted z-score on the final would be $0.5 * -0.52 = -0.26$, and the percentile ranking is roughly $(100\% - 20\%)/2 = 40\%$.
3. The 50th percentile is the mean, and so we would predict the mean, or 50th percentile, on the final.
4. We would predict the mean, or 50th percentile, on the final.

Review Exercises

- See Review_exercises8.pdf
- lecture07_code_updated.R gives the code that I used to create the plots in the exercises and has functions for the following:
  - displaying scatter plots
  - displaying conditional and marginal distributions,
  - calculating and displaying conditional means,
  - displaying graphs of averages, and
  - displaying the regression line and graph of averages on the scatter plot.
- Feel free to use the code in lecture07_code.R to help with the review exercises.